

Softmax

experimental, single fit, evaluated on 2016-2026, measured against the market

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Model identity

Every figure below is read from a declared artifact. This report fits nothing, loads no booster, recomputes no prediction and calls no betting optimiser.

MODEL Softmax experimental	EVALUATION YEARS 11 2016-2026	ENSEMBLE MEMBERS 1 one fit, seed 42
BOOSTING ROUNDS 400 3-fold chronological CV	FEATURE SET production 28 declared inputs	

Softmax is experimental.

It is not the production model. Nothing here is a promotion decision, and the promotion gates it can and cannot satisfy are in the technical appendix.

HOW EACH EVALUATION YEAR WAS FITTED

The 2016 evaluation model was trained on the initial 2010-2015 set. Each later evaluation adds the preceding complete season, so the training window expands by one year; the 2026 model uses 2010-2025. Every evaluation year is fitted separately -- 11 fits in all -- and no evaluation year's outcomes appear in its own fit.

Deltas in this report use `model_minus_market_log_loss: race_multinomial_log_loss`, where **negative** is better. That is stated because three delta fields existed in this repository and they did not all point the same way.

Provenance -- the source commit, the config and data digests, the environment and every input this report read by content digest -- is in the technical appendix and in `run_lock.json` beside this file. Canonical publication fails without it.

Executive summary

MEAN DELTA VS MARKET -0.001087 negative is better	RACES 7,933	YEARS BETTER 7/11
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RACE-LEVEL WIN RATE

52.5%

Softmax is better than the market by 0.001087 race_multinomial_log_loss on 7,933 races. The direction holds in 7 of 11 evaluated years. Race by race it is ahead 52.5% of the time, which is the more honest figure: a mean delta can be carried by a few races.

Evaluation population and market reference

THE MARKET REFERENCE

Every metric below is reported beside the market's own implied probabilities on the same races, normalised within each race, and the delta is this model's race log loss minus the market's on the same races. One yardstick across every published report, so their headline numbers mean the same thing and can be read against one another. The market is a measurement, not a model this one is defined against.

THE EVALUATION POPULATION

Every figure in this report covers the standardized evaluation window, which begins in 2016. One window for every published report, so their headline numbers are computed on the identical race keys and can be compared -- which is checked by digest before anything is published, not asserted in a footnote. 2015 is earlier than the window and is excluded from every evaluation figure below, including the bankroll paths. The feature-effect grid is the one exception, and says so where it appears: a learned response direction is a property of each year's fit rather than of the evaluation population.

TEST RACES

7,933

YEARS

11

2016-2026

Splits present: test, train_apparent. Only the forward test split is scored above. The other exists to be compared against it -- an apparent-fit number looks far better than a forward one -- and is never pooled with it.

Out-of-fold margin sources: model.

One or more years is partial

2026 does not span a full season, so its per-year figures rest on fewer races than the others. Partial years are marked in every table rather than dropped, because dropping them would silently change the evaluation population.

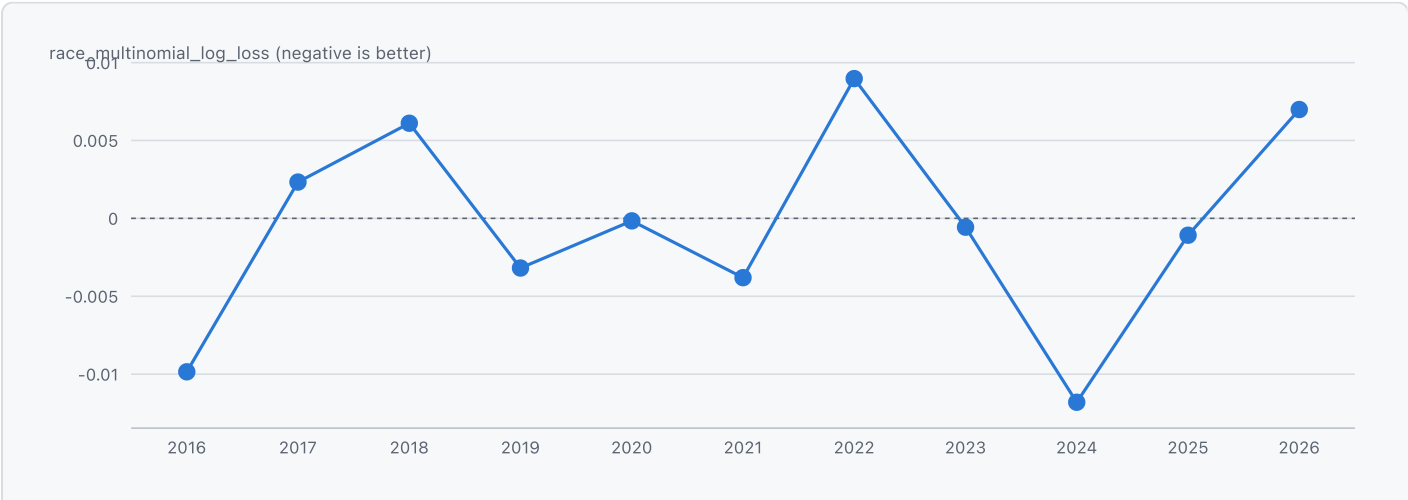
Overall probability performance

races	model log loss	market log loss	model minus market log loss	better race share
7933	2.027576	2.028662	-0.001087	0.525022

One row, pooled over every evaluated race, and measured against the market:
model_minus_market_log_loss is the model's race log loss minus the market's on the same races, so negative means the model is better. The per-race distribution is what the year-by-year section shows; a pooled mean can be carried by a handful of races and this figure alone cannot tell you whether it was.

Year-by-year probability performance

test year	races	model log loss	market log loss	model minus market log loss	better race share	is partial year
2016	744	2.089420	2.099277	-0.009857	0.543011	no
2017	772	2.023892	2.021559	0.002333	0.509067	no
2018	756	2.067610	2.061504	0.006106	0.494709	no
2019	759	1.929638	1.932827	-0.003188	0.528327	no
2020	799	2.010242	2.010400	-0.000158	0.511890	no
2021	783	2.002849	2.006659	-0.003809	0.544061	no
2022	783	2.060168	2.051195	0.008973	0.512133	no
2023	796	2.023505	2.024066	-0.000561	0.531407	no
2024	792	2.013938	2.025744	-0.011806	0.558081	no
2025	748	2.056314	2.057397	-0.001082	0.528075	no
2026	201	2.033219	2.026224	0.006994	0.482587	yes



Mean delta by year Mean delta against the market for each evaluated year. Zero is the market.

▼ Table view

Category	race_multinomial_log_loss (negative is better)
2016	-0.009857
2017	0.002333
2018	0.006106
2019	-0.003188
2020	-0.000158

Category	race_multinomial_log_loss (negative is better)
2021	-0.003809
2022	0.008973
2023	-0.000561
2024	-0.011806
2025	-0.001082
2026	0.006994

Ahead of the market in 7 of 11 years. 1 row(s) are italicised as partial or fallback years, shown rather than dropped so the evaluation population stays visible.

Calibration and sharpness

Reliability by pipeline stage, on bins fixed across the whole report so two stages are comparable, pooled from the per-year counts rather than averaged over years. Stages: market, raw_utility, temperature.

MARKET

Pooled over every test year.

bin	runners	wins	predicted probability	observed frequency	observed / expected
0.00-0.01	10,889	36	0.0062	0.0033	0.538
0.01-0.03	16,603	263	0.0171	0.0158	0.925
0.03-0.05	17,886	625	0.0365	0.0349	0.956
0.05-0.07	15,094	901	0.0625	0.0597	0.955
0.07-0.10	10,318	843	0.0874	0.0817	0.935
0.10-0.15	11,807	1,511	0.1223	0.1280	1.046
0.15-0.20	6,323	1,130	0.1728	0.1787	1.034
0.20-0.30	5,668	1,389	0.2419	0.2451	1.013
0.30-0.50	2,723	1,040	0.3651	0.3819	1.046
0.50-1.00	325	195	0.5566	0.6000	1.078

RAW_UTILITY

Pooled over every test year, for this run's single fit (seed 42).

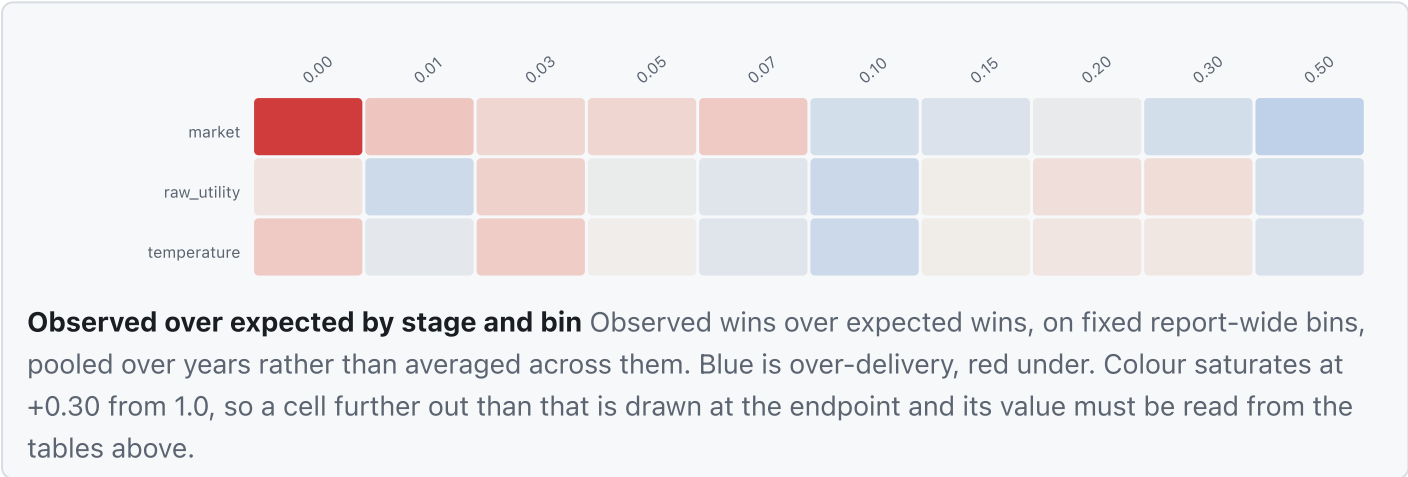
bin	runners	wins	predicted probability	observed frequency	observed / expected
0.00-0.01	13,036	59	0.0046	0.0045	0.978
0.01-0.03	16,113	286	0.0168	0.0177	1.055
0.03-0.05	17,932	650	0.0383	0.0362	0.946
0.05-0.07	16,031	997	0.0616	0.0622	1.010
0.07-0.10	8,623	764	0.0863	0.0886	1.027
0.10-0.15	9,655	1,256	0.1228	0.1301	1.059
0.15-0.20	6,649	1,141	0.1726	0.1716	0.994
0.20-0.30	6,131	1,444	0.2427	0.2355	0.971

bin	runners	wins	predicted probability	observed frequency	observed / expected
0.30-0.50	3,050	1,095	0.3708	0.3590	0.968
0.50-1.00	416	241	0.5557	0.5793	1.043

TEMPERATURE

Pooled over every test year, for this run's single fit (seed 42).

bin	runners	wins	predicted probability	observed frequency	observed / expected
0.00-0.01	12,747	56	0.0047	0.0044	0.935
0.01-0.03	16,119	279	0.0170	0.0173	1.019
0.03-0.05	17,765	639	0.0384	0.0360	0.937
0.05-0.07	16,255	998	0.0617	0.0614	0.996
0.07-0.10	8,788	779	0.0863	0.0886	1.027
0.10-0.15	9,788	1,269	0.1228	0.1296	1.056
0.15-0.20	6,694	1,149	0.1725	0.1716	0.995
0.20-0.30	6,121	1,457	0.2424	0.2380	0.982
0.30-0.50	2,968	1,082	0.3706	0.3646	0.984
0.50-1.00	391	225	0.5560	0.5754	1.035



The largest departure is the 0.000 to 0.010 bin, and every stage misses there: market 0.54, raw_utility 0.98, temperature 0.94. market is furthest from 1.0 at 0.54, meaning that bin won fewer races than it priced; raw_utility is closest at 0.98, so this bin is where the pipeline moves calibration most. On the market stage that bin holds 10,889 runners, the 6th smallest of 10 bins, so the departure is not a thin-bin artefact.

SHARPNESS

stage	races	mean entropy	mean normalised entropy	mean top probability	mean field size
market	7,933	2.0688	0.8274	0.2907	12.3076
raw_utility	7,933	2.0227	0.8093	0.3075	12.3076
temperature	7,933	2.0312	0.8126	0.3045	12.3076

Entropy normalised by field size, so an eight-runner race and a fourteen-runner race are on one scale, and averaged over years weighted by races rather than as a mean of yearly means. A sharper distribution is not

automatically better: it is only better if the calibration above holds.

Top-choice movement

test year	n races	n disagreement races	disagreement share	model top win rate	market top win rate	disagreement mean delta	agreement mean delta
2016	744	99	0.133065	0.222222	0.252525	-0.019522	-0.008373
2017	772	79	0.102332	0.341772	0.177215	-0.029637	0.005978
2018	756	96	0.126984	0.177083	0.135417	0.007805	0.005859
2019	759	68	0.089592	0.220588	0.250000	-0.000335	-0.003469
2020	799	71	0.088861	0.169014	0.197183	0.030911	-0.003188
2021	783	86	0.109834	0.197674	0.244186	0.003447	-0.004705
2022	783	63	0.080460	0.222222	0.238095	0.004219	0.009389
2023	796	52	0.065327	0.076923	0.365385	0.020174	-0.002010
2024	792	69	0.087121	0.173913	0.202899	0.003551	-0.013272
2025	748	69	0.092246	0.144928	0.188406	-0.011842	0.000011
2026	201	14	0.069652	0.142857	0.285714	0.043046	0.004295

How often this model's first pick differs from the market's, and whether it is right when it does. Disagreeing more is not better on its own -- the useful figure is the win rate *conditional* on disagreeing, because a model that disagrees often and loses is worse than one that never disagrees. This run fitted one seed, so there is no member agreement to report beside the disagreement with the market.

Ensemble stability

Not applicable to this model. There is no ensemble to be stable: this run fitted a single seed, so there is no member disagreement to measure. The headline probability is that one fit's own output, which the identity card states.

Feature and correction effects

Not applicable to this model. There is no market correction belonging to this model to attribute. This model learns a utility directly from the feature matrix, in which the market's implied probability is one ordinary column rather than a base margin, so a per-feature attribution here would decompose the utility -- a different quantity from the market correction the attribution artifact declares, and not comparable with the effects reported for the market-anchored models. What this model's features do to its probabilities is therefore not published rather than published under a name that would mislead.

Betting and bankroll analysis

Not applicable to this model. No bet ledger has been settled for this model. It is the unanchored baseline the offset architecture exists to improve on, and it has not passed a promotion gate; settling it would produce a bankroll path that reads as a recommendation for a model nobody proposes to stake.

Simulation and uncertainty

Not applicable to this model. There is no settled outcome to put an interval on. Both intervals this report can show resample a settlement rather than a prediction -- race days drawn with replacement, and winners drawn from the model's own probabilities and settled at Win Odds -- and no bet ledger has been settled for this model, for the reason the betting section gives.

Technical appendix and provenance

CONFIGURATION

Item	Value
Features	28
Features the booster learns on	28

Reduced dictionary. The per-feature classification, response direction and stability entries are derived from the market-correction attribution, which this run does not produce.

PROVENANCE

Item	Value
Source commit	0422b20c2039
Source tree	dirty at generation
Config digest	sha256:8692cb70f72e43ebfdafc9aeab897531825427df19c5b68c13f9f4b171fe9580
Identity source	artifact
Model view	softmax_member
Artifact root	results/softmax_report
Inputs read	7 artifact(s), locked by digest in run_lock.json
Evaluation race keys	7933 races, sha256:342d95ee674808a0d7ae4d0df5b0f958da1898863e25c2898bb71d9f58392e58
Environment: numpy	2.5.2
Environment: pandas	3.0.5
Environment: python	3.12.11
Environment: xgboost	3.4.1

The generating source tree was dirty

This run's provenance records uncommitted changes, so the commit named above does not fully describe the code that produced these numbers. Treat the figures as indicative until the report is regenerated from a clean tree.

Every input this report read is listed in `run_lock.json` by content digest, together with the lock digest of the run that produced them -- which is what chains a published figure back through the report to the fit and from there to the feature table's own digest.

Limitations and the retrospective-odds warning

Retrospective odds, restated

Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

This report publishes no stake, return or bankroll figure -- see the betting section for why -- so nothing here rests on that warning. It is restated anyway, because the odds it concerns are the same historical result-page odds every settlement in this repository uses, and a reader arriving here should not conclude the limitation has been lifted.

Every test year is also development data

Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

WHAT ELSE THIS REPORT DOES NOT ESTABLISH

- It measures Softmax against the market on the same races. It does not measure this model against the other published models except through that shared yardstick.
- It publishes no interval, so no figure in it carries a stated sampling range at all -- a single number is not evidence of precision.
- Every figure is out of sample in time but in sample in *design*: the feature set, the hyperparameters were chosen with these years visible.

- Nothing here is a shadow test. A profitability claim needs a locked model run forward against timestamped pre-race prices.

Unverified source state

This run recorded a dirty source tree, so its numbers cannot be reproduced from the named commit alone.